

EXPERIMENT 21

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Problem Statement

Write a menu-driven C program that accepts two positive integers **m** and **n**, where $m \leq n$. Using a switch statement, repeatedly provide options to print multiplication, addition, subtraction, and division tables for every integer from **m** to **n**, or exit. Division results must be displayed to two decimal places. Invalid menu choices must produce an error message.

Algorithm

1. Start.
2. Read **m** and **n**. Validate that both are positive and $m \leq n$; otherwise request them again.
3. Repeat until the user chooses Exit:
 - a. Display the menu: 1-Multiplication, 2-Addition, 3-Subtraction, 4-Division, 5-Exit.
 - b. Read choice.
 - c. Use switch(choice).
 - d. For choices 1 to 4, loop **i** from **m** to **n** and **j** from 1 to 10.
 - Choice 1: print $i * j$.
 - Choice 2: print $i + j$.
 - Choice 3: print $i - j$.
 - Choice 4: print $(\text{double})i / j$ to two decimal places.
 - e. Choice 5: terminate the loop.
 - f. Default: print an invalid-choice error message.
4. Stop.

Test-Case Table

TC	Input / Condition	Expected Result
1	$m=2, n=3$; choice=1	Print first 10 multiples for 2 and 3.
2	$m=2, n=2$; choice=2	Print $2+1$ through $2+10$ (3 through 12).
3	$m=1, n=2$; choice=3	Print subtraction tables using 1 through 10.
4	$m=2, n=2$; choice=4	Print $2/1=2.00 \dots 2/10=0.20$.
5	$m=1, n=1$; choice=5	Program exits normally.
6	$m=0, n=3$	Reject because m is not positive; request input again.
7	$m=4, n=2$	Reject because $m > n$; request input again.
8	$m=1, n=1$; choice=0	Display invalid menu choice and show menu again.
9	$m=1, n=1$; choice=6	Display invalid menu choice and show menu again.
10	Boundary: $m=1, n=1$; choices 1-4	All four operations work for the smallest valid range.

C Program

```
#include <stdio.h>

int main(void) {
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n ( $m \leq n$ ): ");
        scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Enter positive integers with  $m \leq n$ .\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
```

```

printf("\n--- MENU ---\n");
printf("1. Multiplication tables\n");
printf("2. Addition tables\n");
printf("3. Subtraction tables\n");
printf("4. Division tables\n");
printf("5. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        for (i = m; i <= n; i++) {
            printf("\nMultiplication table of %d:\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d x %d = %d\n", i, j, i * j);
        }
        break;

    case 2:
        for (i = m; i <= n; i++) {
            printf("\nAddition table of %d:\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d + %d = %d\n", i, j, i + j);
        }
        break;

    case 3:
        for (i = m; i <= n; i++) {
            printf("\nSubtraction table of %d:\n", i);

            for (j = 1; j <= 10; j++)
                printf("%d - %d = %d\n", i, j, i - j);
        }
        break;

    case 4:
        for (i = m; i <= n; i++) {
            printf("\nDivision table of %d:\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d / %d = %.2f\n", i, j, (double)i / j);
        }
        break;

    case 5:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid menu choice. Please enter 1 to 5.\n");
}
} while (choice != 5);

return 0;
}

```

Sample Execution

Input: m = 2, n = 2, choice = 4

Output:

```

2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
...
2 / 10 = 0.20

```

Compilation and Execution

```

gcc experiment21.c -o experiment21
./experiment21

```

Result

The menu-driven C program was prepared using loops and a switch statement. It validates m and n, performs all four requested table operations for every integer from m to n, displays division results to two decimal places, handles invalid menu choices, and continues until Exit is selected.